

GABRIEL ADRIAN MANAS

Data Analyst | Data Scientist

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Profile

Medical physicist and quantitative researcher moving into **data science**, with an MS in Applied Physics and a strong foundation in **Python, statistics, and numerical modeling**. I turn measurement, imaging, and simulation data into rigorous, decision-ready analysis — and communicate it through dashboards in **Power BI and Tableau**.

Technical Skills

Programming

Python, MATLAB, SQL

Analysis & Modeling

pandas, NumPy, SciPy, statistical analysis, numerical methods, curve fitting, Monte Carlo (PHITS)

Visualization & BI

Power BI, Tableau, Matplotlib, Excel

Domain Data

Imaging QA/QC, dosimetry, radiation measurement

Tools

Jupyter, Git, LaTeX

Education

M.S. Applied Physics (Medical Physics)

University of Santo Tomas
2026 – Present

B.S. Applied Physics (Instrumentation)

University of Santo Tomas
2021 – 2025

Thesis: Numerical Evaluation of Bragg–Gray Cavity Integrals — numerical modeling of stopping-power ratios in radiation dosimetry (Python / MATLAB).

Core Strengths

- Statistical & numerical analysis
- Scientific computing in Python
- Data visualization & storytelling
- QA/QC & measurement data

Experience

Junior Medical Physicist

Apr 2026 – Present

Medical Physics & Health Physics Services, Inc. (MPHPS)

Python, Excel, Imaging QA

- Perform **performance testing and quality control** across imaging modalities (CT, radiography/fluoroscopy, mammography, PET/CT, SPECT/CT), capturing quantitative image-quality and radiation-output measurements.
- Analyze measurement data against regulatory tolerances to verify equipment performance and **flag deviations**, working with repeated QC metrics across machines and sites.

Science Teacher (Elementary – Senior High)

Jun 2025 – Apr 2026

Hephzibah School Inc.

- Taught science and quantitative reasoning, translating statistical and analytical concepts into clear, accessible explanations — sharpening data-communication skills.

Medical Physics Intern

May – Jun 2024

UP – Philippine General Hospital (PGH)

Excel, Dose Calculations, QA Documentation

- Collected, encoded, and validated **patient dose data** and performed activity and dose calculations across diagnostic radiology, nuclear medicine, and radiation oncology.
- Conducted radiation surveys and compiled QA/compliance documentation, organizing measurement datasets across X-ray, CT, fluoroscopy, MRI, SPECT/CT, PET/CT, brachytherapy, and LINAC systems.

Research & Data Projects

Undergraduate Thesis — Bragg–Gray Cavity Integrals

2025

Python, MATLAB — numerical analysis

- Built numerical models for stopping-power ratios in radiation dosimetry; **modeled mass electronic stopping power as a Gaussian** and evaluated cavity integrals by numerical integration.

Monte Carlo Simulation Study

2025

PHITS — radiation transport

- Ran Monte Carlo simulations and analyzed simulation output to determine Bragg–Gray stopping-power ratios.

Research Dissemination

2025

Quantitative results — 3 venues

- Presented findings at the **43rd Samahang Pisika ng Pilipinas** Conference, the UST RCNAS Symposium, and the UST Mathematics & Physics Seminar.